On the Possibility of a Morally Sound LLM

This project was originally drawn up by Chengheng Li Chen and Dr. Maite López Sánchez. The original code was developed by Chengheng Li Chen based on an *Unsloth* template which was adapted to study a model's responses to *Reddit*'s repository *Am I the asshole* according to a set of personalities stemming from different political orientations. It was reprised with a slightly new orientation by Berta Fitó Casas, the author of this report, still under the direction of Dr. López Sánchez, to provide a preliminary overview on the possibility of training an LLM (Large Language Model) so it follows certain moral precepts when faced with dilemmas a person might encounter in their day-to-day life, taken from the same *Reddit* repository. For that purpose, three aspects ought to be defined before starting to develop it: **laws** (section 1), **sanctions** (section 2), and **schools of thought** (section 3).

1 Laws

Given that the test subject is an Artificial Intelligence model, intent or complex, organically constructed moral frameworks are not to be expected. The training will need hard indications on what behavior the model is to engage in when faced with certain dilemmas. For this reason, the project will first focus on the different scholarly proposals on how to engage with moral laws.

Laws can be classified in three main ways: by their **origin**, by their **scope**, and by their **capacity**. When looking at their origin, laws may be seen as **divine** (originating from a deity or supernatural being) or **human** (created by human beings). Depending on their scope, laws may be **natural** (physical axioms), **positive** (regulations of an individual's actions within a community), or **moral** (internal, conscience-based regulations). Finally, a law may be **imperative** (it commands, although it is possible to disobey) or **impositive** (it is impossible not to follow).

Bellow, *Figure 1* arranges the classifications in a schematic form.

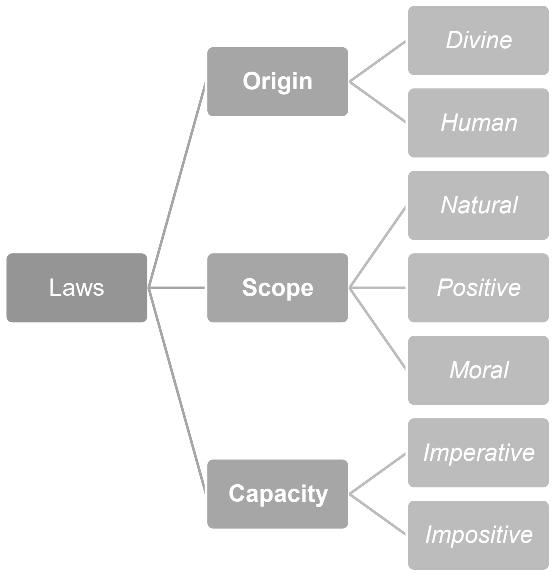

Figure 1. Classification of laws

Laws of a divine origin will not be observed, as it would require choosing one specific religion or finding a set of universal directives common to most doctrines.

2 Sanctions

A law entails a sanction, which is the act through which a competent authority ratifies and validates a law. Sanctions may be classified depending on whether they are external or internal to the individual.

External sanctions may be natural (a consequence of the act without the intervention of an agent), social (a positive or negative valuation from the community), or legal (a sentence applied according to the law). **Internal sanctions** are considered negative (shame, remorse, regret) or positive (joy, moral satisfaction). Internal sanctions will not be dealt with in this project, as an Artificial Intelligence model does not have the capability to experience the internal states required to feel a sanction resulting from an automated action.

Bellow, *Figure 2* arranges the classification in a schematic form.

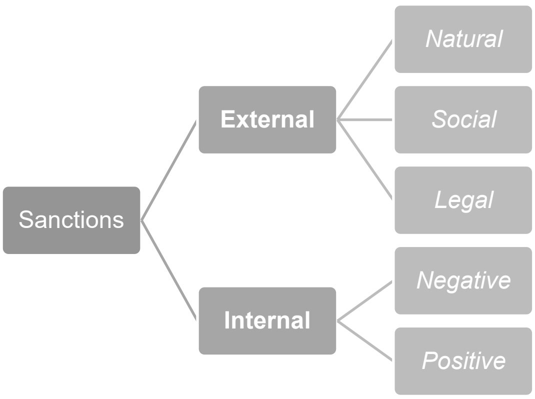

Figure 2. Classification of sanctions

Internal sanctions will not be dealt with in this project, as an Artificial Intelligence model does not have the capability to experience the internal states that would lead to *feeling* a sanction resulting from an automatised action.

3 Schools of Thought

The project will observe five main schools of thought on how to deal with **moral laws**. They disagree on the laws' **origin**, **characteristics** and the way through which they are **known**. They will be broadly evaluated according to the feasibility of their implementation into the training of an Artificial Intelligence model.

- **Moral absolutism** defends that moral laws are universal (in that they apply to every human being, everywhere and anytime) and objective (so they are independent from any consideration, it being from a time, a community, etc.). It can also be interpreted as defending that moral laws do not admit any kind of exception or infraction. It opposes moral objectivism. Immanuel Kant's categorical imperative is an example of moral absolutism.

- **Moral objectivism** tackles the possibility of moral conflict, where two or more moral laws might force disobeying the others to comply with them, which leads it to apply a hierarchical organization to solve these cases. It defends there are *prima facie*[1] moral laws, that is, that they are admitted without need for reason as it is clearly intuitive that they are moral laws. They are intuitive because they enable the development of humanity and community and assume all human beings share the same nature, and so they have the same interests and necessities. The validity of moral laws resides in their intent to safeguard these interests and necessities and may be thought to originate either from reason (natural moral laws) or a deity. It opposes moral absolutism.

  A proposed ranking of moral laws follows, from most binding to least so.

  1. To not kill innocent people.
  2. To not cause unnecessary harm nor suffering.
  3. To not deceive nor steal.
  4. To comply with promises and honour one's agreements.
  5. To not deprive anyone from freedom.
  6. To act justly.
  7. To be truthful.
  8. To help others.
  9. To be grateful.
  10. To obey those positive laws that are just.

  An example of such doctrine may be found in Saint Thomas of Aquino's law of not killing, which he allows breaking in the specific case of a tyrant oppressing a people.

- **Moral realism** binds all events with moral valuations and separates the latter from everyone's judgement. Such valuations are fostered by the event, not the individual, whose only job is to announce them. Such school of thought would consider that, as an example, physical suffering is the negative moral valuation of an event. It opposes all other doctrines which would consider such a product a neutral fact, without any intrinsic morality. Baruch Spinoza, who derived his ethical principles from metaphysics, is a moral realist to some extent.

- **Moral relativism** acts on the interpretability of a law to lead to different, technically correct subjective manifestation of its formulation. It argues that it may also be the case that the law is bound the circumstances of its application. It opposes moral absolutism,

[1] Meaning *at first glance*.

as it defends that morality is not universal and, alternatively, holds that moral laws are dependent on its time, culture, or group. Moral laws are, in fact, validated by the group. Such relativism forbids different groups from judging the moral laws of others, which severely complicates the coexistence in plural societies.

- **Moral subjectivism** is an extreme form of moral relativism. Just as moral relativism did, it opposes moral absolutism. It argues that the subject's judgement is the foundation or limit for morality. It is grounded on affection and coherence, as it is the consistency between what the subject does and their personal values what gives their morality validity. Jean-Paul Sartre would be a representative of moral subjectivism. **Moral nihilism** an extreme form of moral subjectivism. It denies the existence of objective moral truths. Although it allows for altruism or willingness to do no harm to others, it does so without believing in said truths.

4 Action Plan

In this section, a preliminary plan to develop the model will be expanded on.

Even while avoiding explicitly unique religious stances in not considering divine laws, **human-made laws may vary greatly** depending on the time and place one might be in. It is the aim of this project to provide a schema of beliefs that is **as universal as possible**, but a very thorough revision would be warranted to make sure any historical o cultural remnant of bias from the authors (or the dataset) is scrubbed out. The short timeframe allotted to it will not allow for such revision, so this shall be considered a very broad, possibly skewed, expression of the ultimate objective of this project.

All three categories of laws, according to their **scope**, should be considered. **Natural laws** are to be assumed, as the relevance of the solutions provided by the model hinge on them being executable in the world. Ethics is a field of study prominently focused on its practical application, rather than theoretical elucubration. The quotidian focus of the present project reinforces this direction. **Positive laws** are to be enforced. It is unknown what use might the imagined model have in the long run, but it is unlikely that it would be beneficial that it could be prone or even allowed to offer solutions that could lead to the user breaking the law to accommodate them. Of course, **moral objectivism** might allow for it as a last resort, but this kind of judgement should not be made by the model, rather the user that would receive the best possible solution without resorting to such extremes. Liability is a difficult topic in the field of Artificial Intelligence. Finally, **moral laws** are a given, considering the objective of the project to train a model to provide solutions for quotidian moral dilemmas.

In this project it will be assumed possible to train a model so that it shows a modicum of nuance in evaluating whether to **break an imperative law**. As previously stated, the model should not incentivise breaking a positive law, but it might be tasked with prioritising a moral law above another, thus breaking or overlooking the one that is deemed less relevant. Conflicts such as this one can happen, for example, in moral objectivism.

While **internal sanctions** have been put aside in this project, **external sanctions** will be considered relevant to the resolution of each case. **Natural** sanctions are to be assumed, just as natural laws were. The consequences they entail are unavoidable and ought to never be overlooked when providing a solution to a dilemma, as physical safety could be hindered, both the individual's (moral agent) or anyone else involved. **Legal** sanctions are the reason positive laws should not be broken, nor should illegal actions be incentivised. They must be considered before providing the final solution to the dilemma. Finally, **social** sanctions are important to prevent social unrest within the community the user will likely be a part of and considering the dilemmas will be quotidian scenes a person might encounter and solve organically.

After the preliminary investigation, **moral subjectivism**, which is based on the individual's moral judgement, has been ruled out as an applicable school of thought. As previously stated, an Artificial Intelligence model does not have intention nor the necessary functions to be able to judge and create an individual moral system.

Most moral trends could be spun to fit the requirements of this project. An example of a school of thought able to go both ways is **moral realism**. Its emphasis on values complicates its implementation, as a complete ontology for them would be required to adequately quantify the model's assessments. However, once such ontology was developed, some merit could be found to assigning moral value to a standardised quantification of suffering, for instance.

This initial assessment favours **moral objectivism** as the most promising out of the present schools of thought. It allows for a clear and structured set of instructions to give to the model and many opportunities for experimentation. An interesting notion to explore could be asking how fitting it is to think of such moral trend as *moral objectivism* when implementing it to an Artificial Intelligence model, to which an instruction is given: *follow ten moral rules, prioritising the lower-numbered ones*. As this is an absolute rule, it can be a worthwhile line of thought to see if this constitutes **moral absolutism**.

4.1 Human Acts and Liability

Moreover, to accurately assess the dilemmas, the model must distinguish between **Acts of a Human** (biological functions) and **Human Acts** (done with knowledge and will). Specifically, the model must be trained to recognize:

1. **Acts of Ignorance**, where an agent wilfully enters a state of not-knowing (like intoxication). In these cases, the agent remains liable.

2. **Mixed Acts**, which are actions taken under coercion or threat. The model must evaluate whether the action would have occurred without such external pressure.

4.2 Limitations

As previously stated, the **timespan** for this project is limited to a few months. This limits its scope, given that it will not be possible to make a thorough assessment of all possible moral implementations. It also shortens the available time to provide a study for the results of the implementations that will be tried out.

One must never forget that the model being used is a pre-trained model, with its behaviour filtered out for our purposes. This model is, in turn, a **product**. As such, there are certain concerns for liability that have already been addressed by the provider of such service. The decisions made during the actual training of the model are beyond the scope of this project, albeit it might profoundly affect its results. We cannot know what parameters or hardcoded considerations were implemented during its initial setup, but it is most probable that some provisions in the way it answers and analyses problems, as well how it interacts with positive laws or the rules of society, have been set prior to our specific prompts.

5 Implementation

This project was developed in a Jupyter notebook, in which a two-stage fine-tuning pipeline for automated reasoning, applied to the task of binary ethical judgement, was implemented. The notebook has been annotated to clarify the code's stages, alongside whichever clarification was deemed necessary.

First, a 4-billion-parameter causal language model (named *Qwen3-4B*) is fine-tuned to evaluate short narrative dilemmas drawn from the *Am I The Asshole* (AITA) Reddit community. Its objective is to classify the author's behaviour as morally *right* or *wrong* according to certain moral axioms. The following *implementations* would have dictated said axioms. As of finishing this project, only the *first* were tried out: a fixed, priority-ordered set of ten deontological axioms based on moral objectivism.

The pipeline proceeds in two stages. In the first one, Supervised Fine-Tuning (SFT) is applied on 100 curated examples. Each example is comprised of a written dilemma paired with a pre-generated chain-of-thought reasoning trace, which is structured as an initial ethical analysis, a dialectical challenge, and a synthesis. This is then followed by a final verdict. This teaches the model the expected output format and exposes it to high-quality reasoning patterns.

In the second stage, Group Relative Policy Optimisation (GRPO) is applied on a balanced set of 1000 examples to refine the model's reasoning through reinforcement learning. Four reward functions guide the process: two structural rewards that check for correct output formatting, one that compares the extracted judgement against the ground-truth label, and a reasoning-quality reward provided by an external teacher model. The teacher model is *Meta-Llama-3-8B-Instruct* as of finishing this project. The teacher model scores the *student* model's (*Qwen3-4B*) reasoning across six dimensions: neutrality, evidence integration, hypothesis testing, logical coherence, nuance recognition, and decision alignment. It finally returns a composite reward signal.

5.1 First implementation

The first implementation to be applied to the model revolves around **moral objectivism**. Specifically, the model is initially given the following setup for its responses:

```
reasoning_start = "<think>"
reasoning_end   = "</think>"
solution_start  = "<O>"
solution_end    = "</O>"
system_prompt =   f"""You are a helpful assistant.
You are given a situation needing a moral solution.
You must reason a moral solution and determine whether the narrator is behaving correctly according to moral reasoning.
You must take into consideration these moral axioms, which are to be prioritised from top (1), with maximum priority, to bottom (10):
1.	To not kill innocent people.
2.	To not cause unnecessary harm nor suffering.
3.	To not deceive nor steal.
4.	To comply with promises and honour one's agreements.
5.	To not deprive anyone from freedom.
6.	To act justly.
7.	To be truthful.
8.	To help others.
9.	To be grateful.
10.	To obey positive laws that are just.
You must accommodate as many axioms as possible and justify each axiom you do not.
Then determine whether the narrator is behaving correctly according to this moral reasoning.
Place your reasoning between {reasoning_start} and {reasoning_end}.
Then, provide your answer "RIGHT" or "WRONG" within {solution_start} and {solution_end} as the final verdict.
"""
```

The final setup was changed, mainly, to shorten it and make it more efficient. It read as follows:

```
"""You are a helpful assistant.
You are given a dilemma. You must decide if the author is acting correctly according to moral reasoning.
You must consider these moral axioms, from top priority (1) to least (10):
1.	To not kill innocent people.
2.	To not cause unnecessary harm nor suffering.
3.	To not deceive nor steal.
4.	To comply with promises and honour one's agreements.
5.	To not deprive anyone from freedom.
6.	To act justly.
7.	To be truthful.
8.	To help others.
9.	To be grateful.
10.	To obey positive laws that are just.
You must accommodate as many axioms as possible. Answer 'No: ' and the numbers of the unapliccable axioms.
Decide if the author is acting correctly in this moral reasoning in 100 characters or less.
Place your answer between {reasoning_start} and {reasoning_end}.
Then answer "RIGHT" or "WRONG" within {solution_start} and {solution_end} as the final verdict.
"""
```

This prompt seemed to allow the model to answer to the dilemmas in a manner that generally befit the annotated solutions in the dataset. However, the results of this experiment are inconclusive due to technical issues in the development of the code: specifically, the teacher model was never fully operational. This may be due to many possibilities: the code had been designed for a different environment (and a slightly different purpose), some libraries had been updated between the end of the prior project and the start of this one and were now in different packages (which was a minor problem) or their compatibilities needed to be revised. The new environment was not as powerful as the prior one, and the models ultimately used for this project needed to be changed, both due to lack of accessibility. Ultimately, most issues were solved, albeit in a much longer timeline than initially expected, but one: the new teacher model (*Hugging Face*'s Llama 3) was never fully functional.

During the solving all other technical issues, Google's *Gemini 1.5 Flash* was used to try out the teacher model's setup. The model, in its free tier, was never capable to hold the teacher model's training, as the tier did not allow for so many tokens to be used up. But it was a good test to see whether the code could reach the training phase. Ultimately, *Hugging Face*'s Llama 3 was chosen as the basis for the teacher model. This forced an unexpected amount of code refactoring, which arose new technical problems to be solved, mainly in the front of library compatibilities. It was very late in the project's timeframe when the code was back to reaching the teacher model's training instruction. However, it was quickly apparent that the model hallucinated spectacularly (at some point it started generating its responses in Dutch and seemed be only capable of contextualizing them around sex; in another trial, it switched to Chinese and started to compose fictional market analysis and complimenting itself). The issue seemed to be that the messages' formatting was incorrect. Although some attempts to fix this, the project came to a close before they could be properly tested.

5.2 Second implementation

A possible second implementation, which is purely theoretical, could be built around duty ethics, which are best represented by Kantian ethics. It would ignore the agent's situation or the consequences of an act, while making the moral subject strictly equal for everyone. Then, an act would only be *good* if it was performed uniquely for the sake of complying with the moral law.

5.3 Third Implementation

Furthermore, it could be interesting to explore a possible implementation based on the Ethics of Care, which is a branch of ethics values that highlights vulnerability and interrelation. The primary virtue is *benevolence* (that is, wanting the good of others). It acknowledges two types of valid reasoning:

1. **Rational/Argumentative**, which is the traditional logical moral reasoning.
2. **Personal/Emotional**, the intuitive reasoning that appeals to subjective feelings and specific relationships.

The Heinz Dilemma was suggested by Carol Gilligan to illustrate the difference in moral thinking between girls and boys. It presented a situation in which a doctor has discovered a medicine that cures a type of cancer. The doctor sells it a very high price. A man called Heinz is married and his wife is ill from the cancer the doctor's medicine can cure, but he is unable to buy it due to its prohibitive price. Gilligan asked whether Heinz should steal it and, according to her findings, the boy favoured stealing it while the girl did not. It might be interesting to implement a set of instructions for the model to follow which would result in each answer (boy-like and girl-like, if we are following Gilligan's classification), and then observe how it would solve the other dilemmas.

5.4 Fourth Implementation

Another experiment could explore the Dialogic or Communicative Ethics of Jürgen Habermas. It might also be referred to as Procedural Ethics and would focus on the *procedure* of reaching a norm through consensus. A few directives for a valid norm for the model could be:

- The norm must solve the **immediate problem** at hand (the dilemma it has been given).
- The norm must be articulated in plain, **understandable terms** for all affected parties (all agents involved in the dilemma).
- The norm must remain **subject to revision** over time (in other applicable dilemmas).